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324412 (24)

BE (4th Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical, EEE

ELECTRO MAGNETIC FIELDS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) Attempt any two parts from the remaining part of the question.

Q. 1. (a) Give the position vectors in cartesian, cylindrical and spherical co-ordinate system. 2

(b) The surface r = 2 and 4, θ = 30° and 50° and ϕ = 20° and 60° indentify a closed surface. Find 7

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(i) the enclosed volume

(ii) the total area of the enclosing sur

(iii) the total length of the twelve edg Q.

the surface.

(c) Find $\bar{E}$ at origin if the following d distributions are present in free space: charge 12 nC at P(2, 0.6), uniform charge density 3 nC/m at x = –2, y = 1 uniform surface charge density 0.2 nC/ x = 2.

(d) Given the electric flux density D = 0.3 nC/m² in free space. Find

(i) E at point P. (r = 2, θ = 25°, ϕ = 3

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(ii) The total charge within the sphere r = 3

(iii) The total electrical flux leaving the sphere r = 4. 7

Q. 2. (a) Define Dielectric strength. 2

(b) Let a uniform surface charge density of 5 nC/m^2 be present at the z = 0 plane, a uniform line charge density of 8 nC/m be located at x = 0, z = 4 and a point charge of 2 μC be present at P(2, 0, 0). If V = 0 at M(0, 0, 5), find V at N(1, 2, 3). 7

(c) Two perfect dielectrics have relative permittivities $\epsilon_{r_1} = 2$ and $\epsilon_{r_2} = 8$. The planner interface between them is the surface

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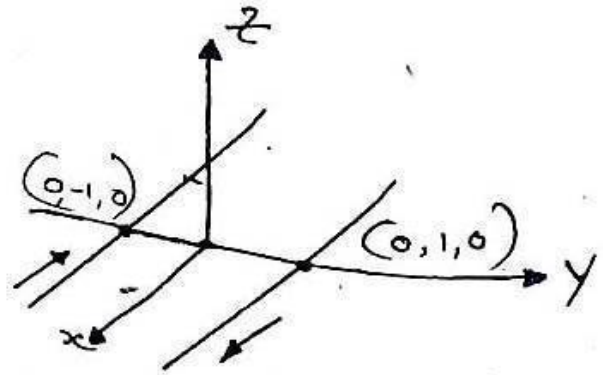
x – 4 + 2z = 5. The origin lies in region 1

$\bar{E}_1 = 100\,\overline{ax} + 200\,\overline{ay} - 50\,\overline{az}$ V/m. Find $\bar{E}_2$

(d) Derive the expression for an electric field due to electric dipole. 7

Q. 3. (a) State amperes circuital law. 2

(b) The parallel filamentary conductors as shown in fig. below lie in free space. Plot |H| versus Y, –4 < Y < 4, along the line x = 0, z = 2. 7



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(c) Let $\overline{A} = (3y - z)\overline{ax} + 2xz\overline{ay}$ Wb/m in certain region of free space (i) show that $\overline{V}.\overline{A} = 0$ (ii) At P(2, –1, 3), find $\overline{A}$, $\overline{B}$, $\overline{H}$ and $\overline{J}$. 7

(d) Evaluate both sides of Stoke's theorem for the field $\overline{G} = 10 \sin\theta\ \overline{a\phi}$ and the surface r = 3, $0 \le \theta \le 90°$, $0 \le \phi \le 90°$. Let the surface have the $\overrightarrow{ar}$ direction. 7

Q. 4. (a) What is magnetization. 2

(b) Given a material for which $X_m = 3.1$ and within which $\overline{B} = 0.4y\overline{az}$. Find (i) $\overline{H}$, (ii) μ, (iii) μ_r, (iv) $\overline{M}$, (v) $\overline{J}$ and (vi) $\overline{J_T}$. 7

(c) Derive the expression for tangential and normal component of boundary condition. 7

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(d) Derive the expression for force and torque on closed circuit. 7

Q. 5. (a) State why Maxwell equations are represented in two different forms. 2

(b) Let $\mu = 3 \times 10^{-5}$ H/m, $\in = 1.2 \times 10^{-10}$ F/m and $\sigma = 0$ everywhere. If $\overline{H} = 2\cos(10^{10}t - \beta x)\overline{az}$ A/m. Use Maxwell's equation to obtain expression for $\overline{B}$, $\overline{D}$, $\overline{E}$ and β. 7

(c) In figure below $\overline{B} = 0.2 \cos 120\pi t$ Tesla and assume that the conductor joining the two ends of the resistor is perfect. It may be assumed that the magnetic field produced

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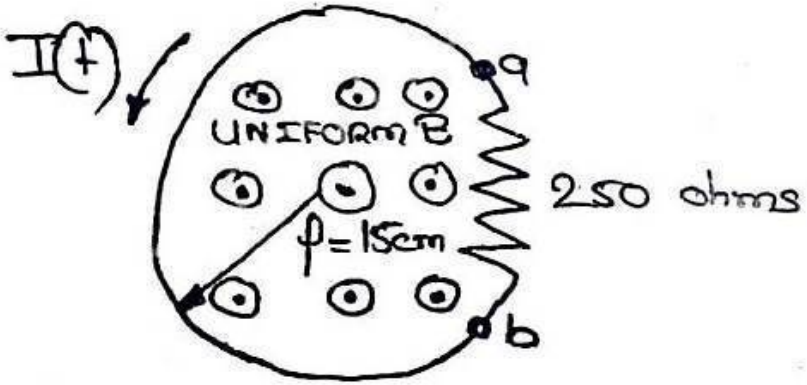
by I(t) is negligible. Find (i) Vab(t) and (ii) I(t). 7



(d) Describe the relationship between field theory and circuit theory. 7

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